

Global Strategy: Class #7

Climate Change, Inclusive Innovation, and Stakeholder Management

Natura Co.

Leandro “Leo” Pongeluppe

The Wharton School – Management Department

pongelup@wharton.upenn.edu / www.LeandroPongeluppe.com

TA: Liz Sandoval Siman

ems6@wharton.upenn.edu



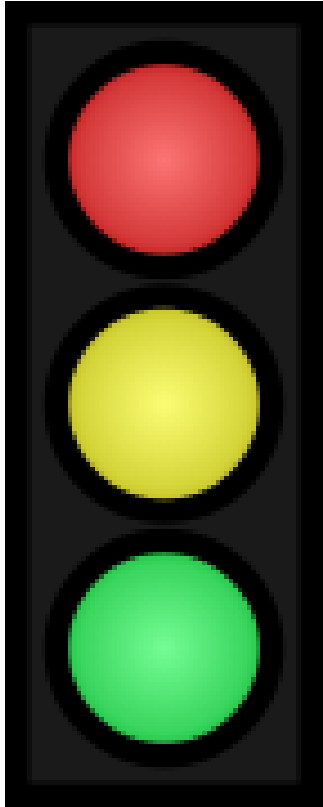
Last Class

- How should we go? **Transaction Cost Economics (TCE) Framework.**
 - Country risk
 - Political Risk might shift governance curves given the increase in contractual hazards associated with “asset expropriation” risk.
 - Institutional Voids and Business Groups
 - Institutional voids impose difficulties in operations in emerging markets. Firms can act in a countercyclical way.
 - At the same time, institutional voids foster the development of business groups, which act to overcome these voids and reach a competitive position internally and internationally.

TODAY

- How should we go? **Stakeholder Management Framework**
 - Inclusive Innovation, Climate change, and Stakeholder Management

What We Are Trying to Solve in the Global Module

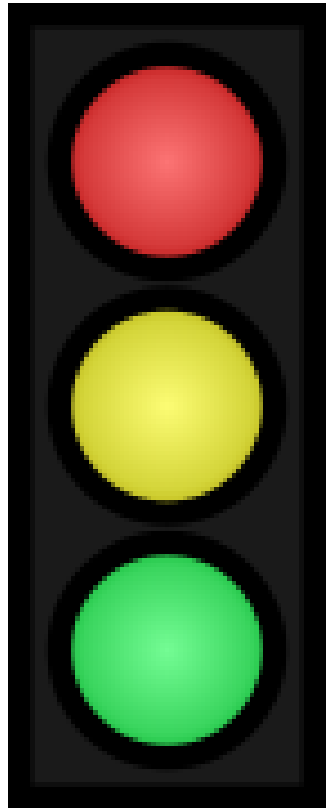


Should we go abroad?

Where should we go?

How should we go?

What We Are Trying to Solve in the Global Module



Should we go abroad?

- **ADDING Value**
- **CIA:** Comparative Institutional Analysis

- Ferrero Group
- Corruption in International Business

Where should we go?

- **CAGE:** Culture, Administrative, Geographic, and Economic
- **AAA:** Adaptation, Aggregation, and Arbitrage

- European Super League
- Apple
- Zara

How should we go?

- **TCE:** Transaction Cost Economics
- **SM:** Stakeholders Management

- Walmart
- McDonald's
- Tata Group
- Natura



Can MNCs help solve global socio-environmental problems?

Narsalay, Pongeluppe, & Light (2015)
“The Hidden Pitfalls of Inclusive Innovation,”
Stanford Social Innovation Review. Winter: 48-53

“Inclusive Innovation” Project

- One billion poor people from emerging markets will rise to the middle class in the next several decades → Huge business opportunity
- Partnership among:
 - Accenture Institute for High Performance
 - Insper (São Paulo, Brazil)
 - Lagos Business School (Lagos, Nigeria)
 - South Asia Initiative at Harvard University (Cambridge, MA, USA)

What are the drivers for MNCs to successfully engage with Bottom-of-the-Pyramid (BoP) populations, generating “Inclusive Innovations”?

Firms and locations

Country	Agriculture	Automobiles	Consumer Goods	Mining	Medical Devices	Telecomm.	Banking
	UNICA & Klabin		Natura	VALE			Itaú
		Shanghai-GM & Vuling	Alibaba & Haier		Siemens	Nokia	
	ESOKO		Promasidor & Nestle				
	ITC	Tata Motors	Hindustan Unilever Limited			Idea	Yes Bank

Inclusive Business Innovations (IBIs) Pitfalls

- Lack of buy-in from top managers – “one strike out”
 - Ex.: **Hindustan Unilever Limited (HUL)** (India) directors defending the idea.
- Focus on the wrong performance metrics
 - Ex.: **ESOKO** (Ghana) creating the infrastructure and training SMS about prices on agricultural goods.
- Failing to recruit talented executives
- Partnering with wrong organizations
- Using old business models
 - Ex.: **Natura** (Brazil) instead works with the communities to create new products and business models.



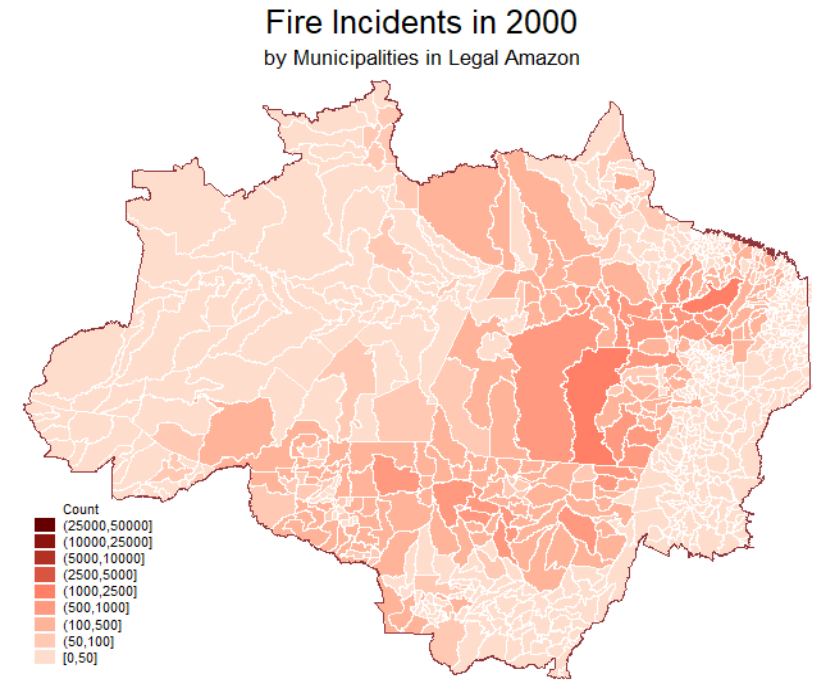
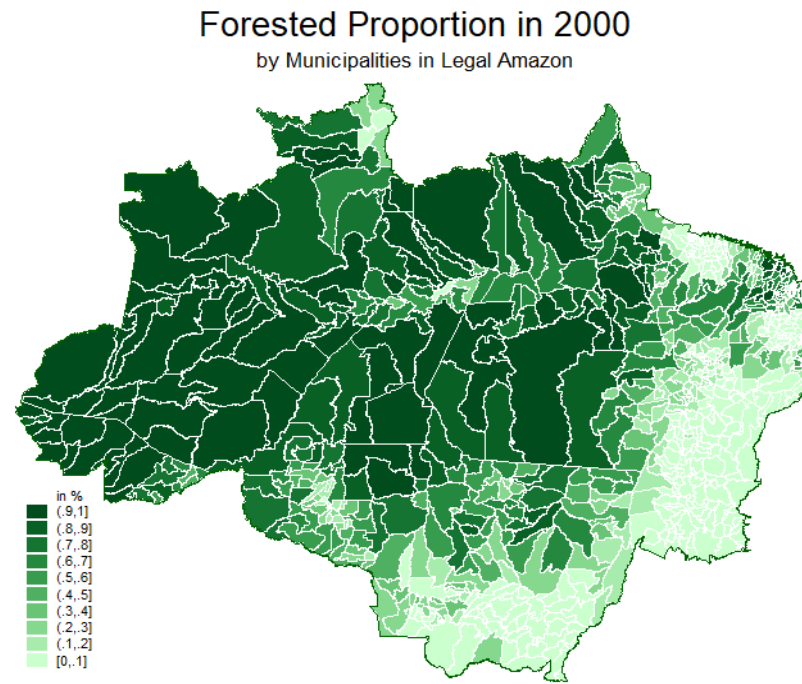
Hindustan Unilever Limited

esoko

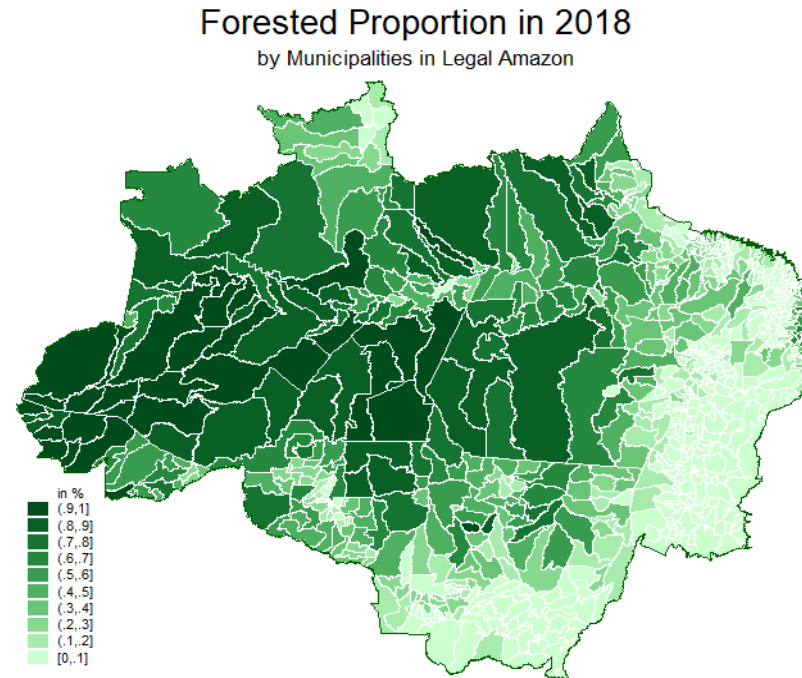


natura

The Forest is Becoming Depleted

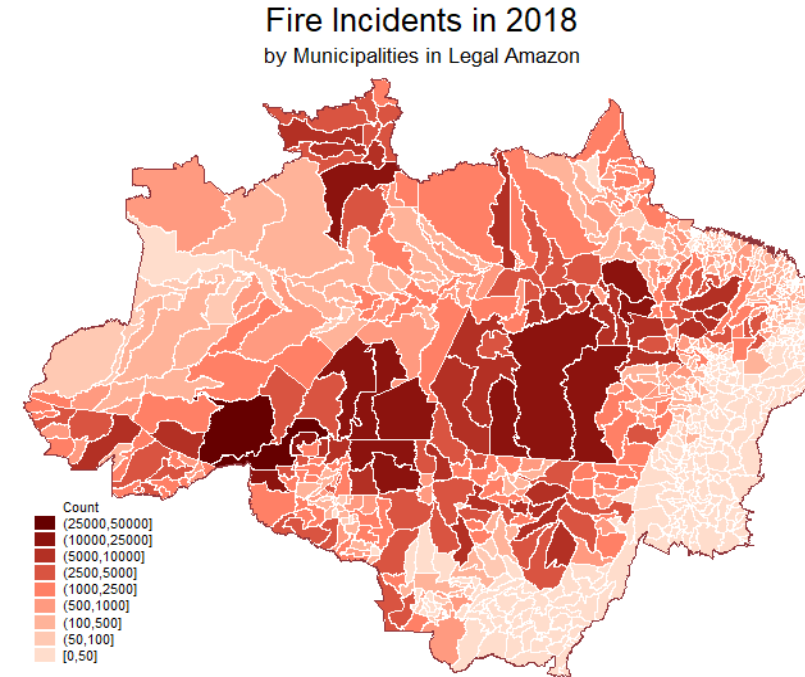


The Forest is Becoming Depleted



Total forest lost area
53.2 million hectares
~ **50 million football fields.**

Nota: data from PRODES and IBGE.



Total fire incidents **increase** from
60 thousand to more than
900 thousands.

Nota: data from PRODES and IBGE.

Can markets, or specifically a firm help to save the Amazon rainforest?

Natura Co. Boehe, Pongeluppe & Lazzarini.
“Natura and the development of a sustainable supply chain in the Amazon region.” Insper Case, 2013.

“Amazon Case Study: Natura’s Supply Chain.”
Americas Quarterly. May 12th, 2021.

Freakonomics Radio, Episode 428 *“The Simple Economics of Saving the Amazon Rain Forest”*.
Click [here](#) to listen.





COOPERATION

Can markets, or specifically a firm help to save the Amazon rainforest?

Natura Co. Boehe, Pongeluppe & Lazzarini. *“Natura and the development of a sustainable supply chain in the Amazon region.”* Insper Case, 2013.



Sources: <https://www.ecohealthalliance.org/2019/08/the-amazon-is-burning>;
<https://www.dtnext.in/News/World/2019/09/05042942/1175373/Explained-Why-the-Amazon-rainforest-is-on-fire.vpf>

In-class Group Exercise, Part I

Each group of students will be randomly assigned to the role of the following **stakeholders**:

1. Natura's shareholders
2. Government
3. FASE (grassroots NGO)
4. Local community (Amazonian suppliers)
5. Customers (of cosmetics)
6. Natura managers

Please discuss in the group (12~15 min) and write one or two paragraphs answering the following question:

What actions would you require from Natura's managers to guarantee that you, as a stakeholder group, are benefiting from its operations in the Amazon?

Natura's shareholders

What actions would you require from Natura to guarantee that **YOU**, as a **stakeholder group**, are benefitting from its operations in the Amazon?

Amazonian Supplier

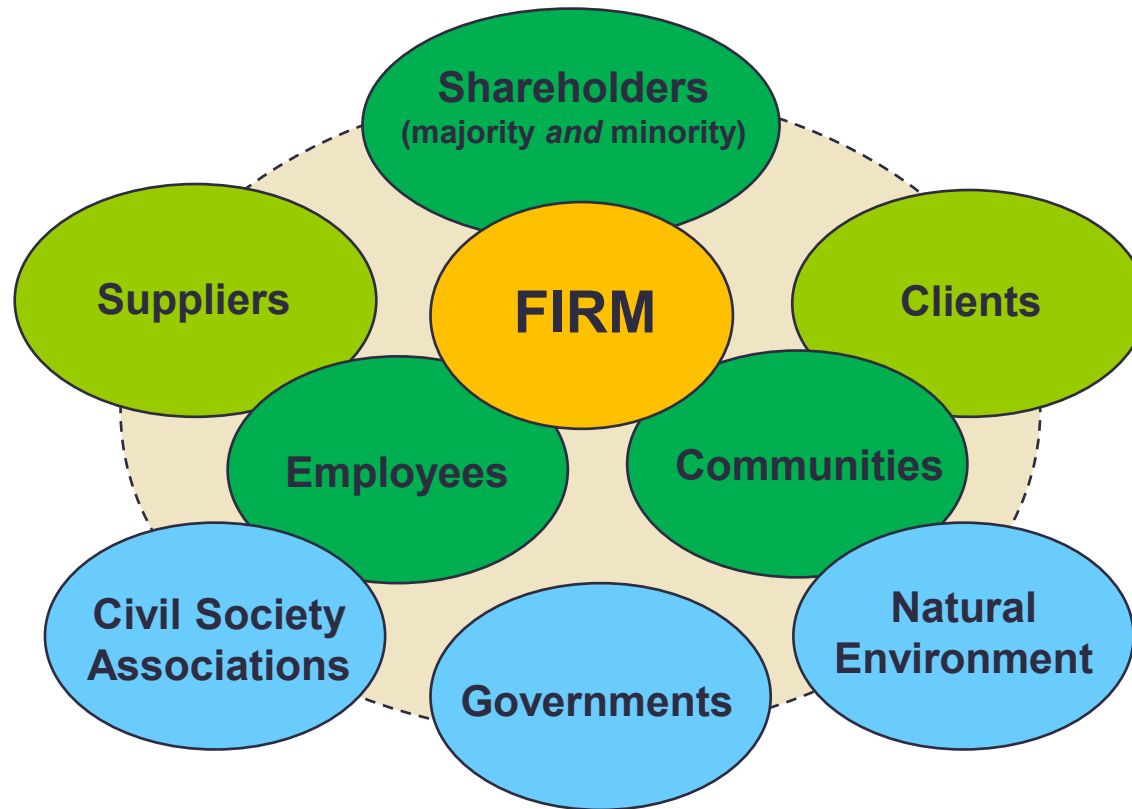


Customers (of cosmetics)

FASE (NGO)

Government

The Stakeholder Management Framework



“Instrumental” stakeholder hypothesis: benefiting stakeholders or avoiding harm can lead to greater profitability in the long run.

Natura

- Founded in 1969 in Brazil. The company operates in **9 countries**, has yearly net revenue of **USD 6.19 billion (2025)**.
- Internationalization greenfield and through Acquisitions. Natura acquired “**Aesop**” (2014-2023), “**The Body Shop**” (2017-2023), and “**Avon**” (2019-2025).
- **UN Champion of the Earth winner** (2015), the world’s largest **B-Corp** (Marquis, 2020).
- Natura sells **cosmetics** incorporating 14 different **Amazonian fruits and nuts** (Gatignon & Capron, 2020).



Flashcards

Natura Completes “Corporate Simplification”: Divests Avon International, Aesop & The Body Shop

Natura completed the sale of Avon International to Regent LP (Dec 2025), sold Avon Russia (Feb 2026), and previously divested Aesop and The Body Shop. The company retains Avon’s Latin American operations. Q4 2025 revenue: R\$6.2B. Full-year recurring EBITDA margin: 14.6%, up 130 basis points YoY. The strategy is clear: **shed global complexity, refocus on Latin America and the Amazon bioeconomy.**

Sources: [Global Cosmetics News \(Dec 2025\)](#), [Cosmetics Business: Avon Russia Sale \(Feb 2026\)](#), [Cosmetics Business: Q4 2025 Results](#), and [Global Cosmetics News: Restructuring Complete \(2025\)](#)

Question: “Natura bought Avon, Aesop, and The Body Shop to go global. Then it sold them all. Using the stakeholder framework, what went wrong? And why does refocusing on Latin America make sense for the firm?”

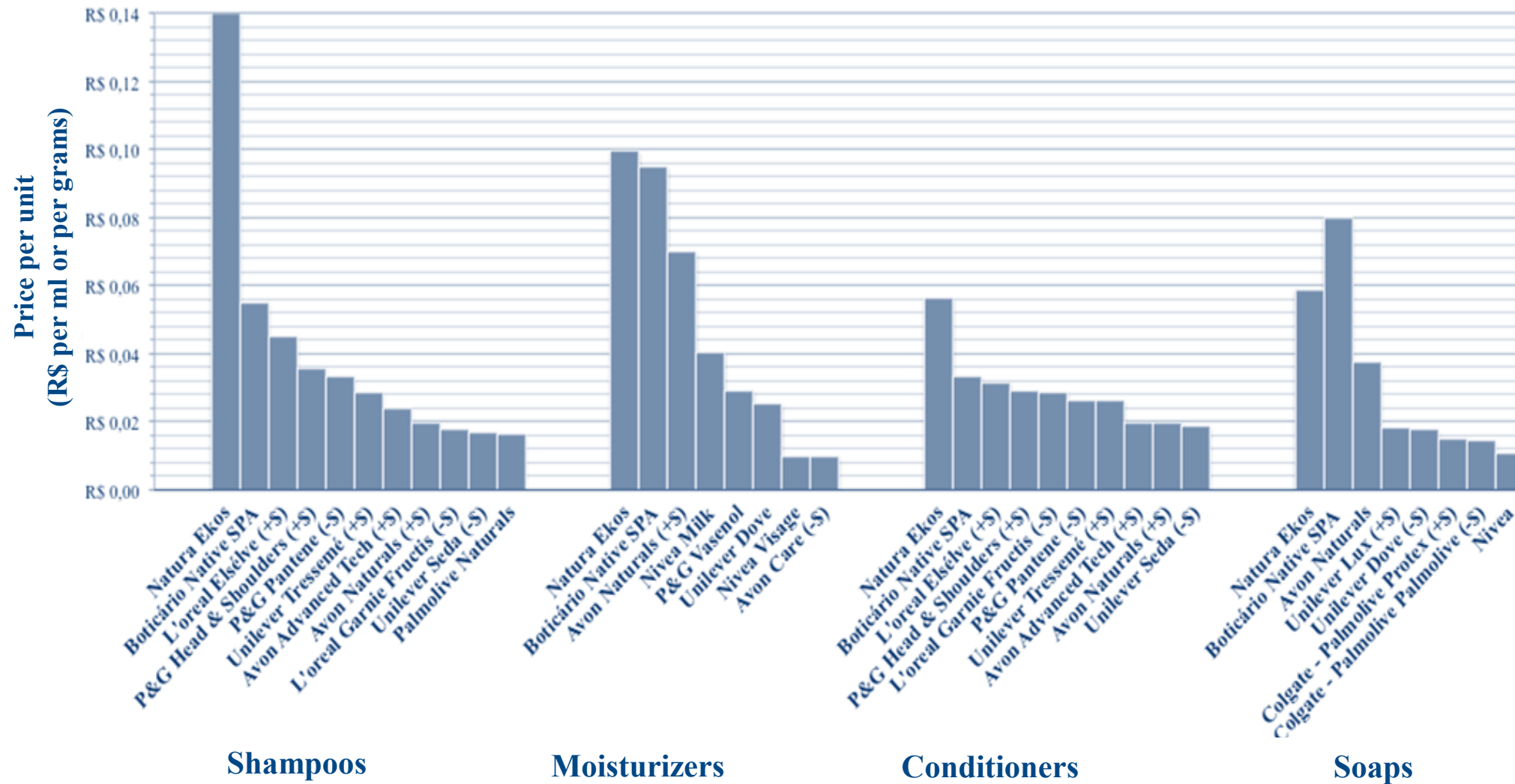
ANALYSIS

Stakeholder Framework, Complexity vs. Alignment: Natura’s acquisitions (Avon, Aesop, Body Shop) expanded its stakeholder map dramatically: new customers in 100+ countries, new regulatory environments, new supply chains, new employee bases. **Managing convergences across this many stakeholders proved unmanageable.** Margins declined; debt increased. **The simplification is an acknowledgment that stakeholder alignment works best when focused.**

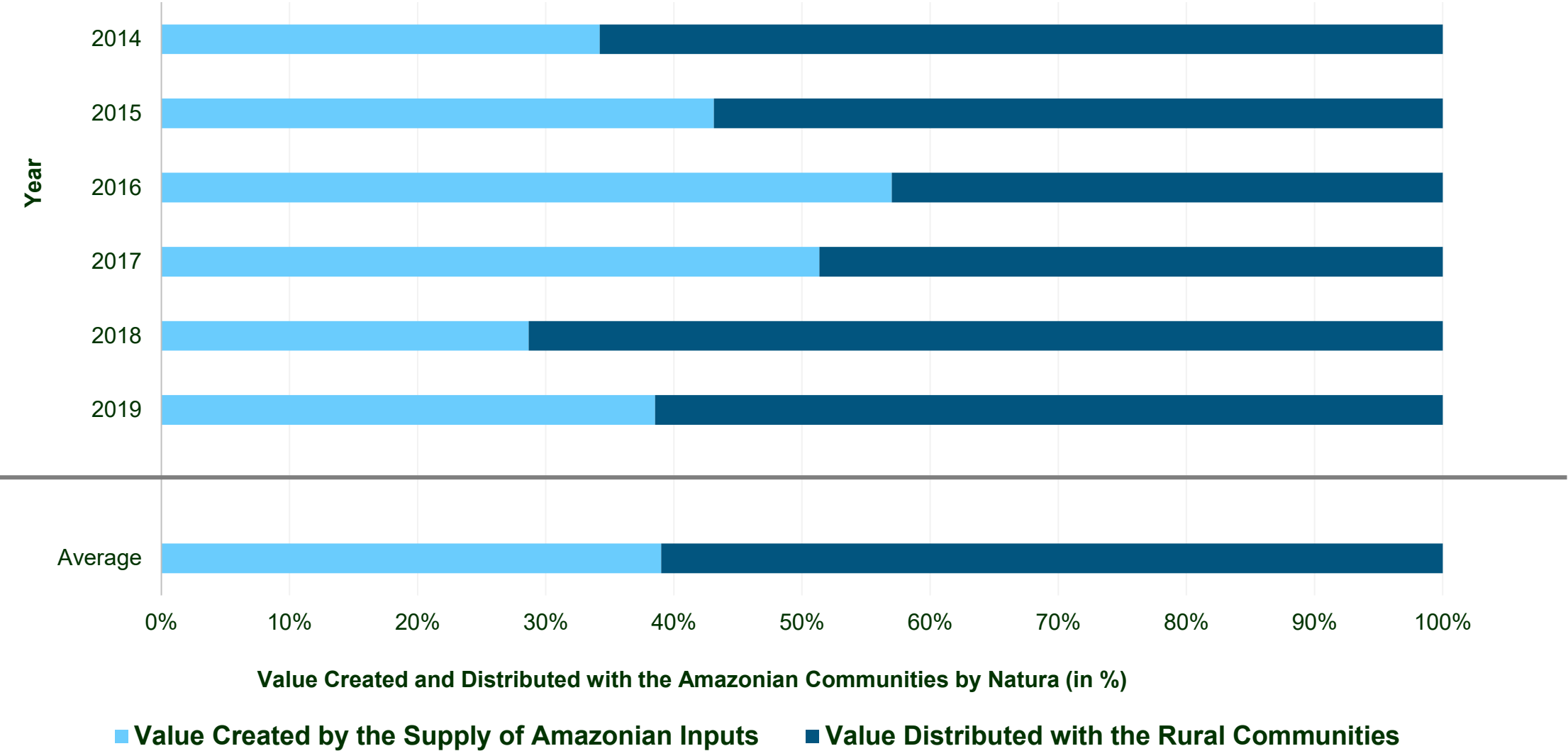
Win-Win Geography: In Latin America, Natura’s stakeholder interests converge naturally: Amazonian communities supply ingredients, Natura provides livelihoods, forests get preserved, consumers get authentic products. This convergence doesn’t exist in Europe or Asia, where Avon’s direct-selling model faces different challenges and the Amazon story has no operational connection.

Profitability as Sustainability Proof: EBITDA margin up 130 bps post-simplification shows that the stakeholder model *is more* profitable when focused.

Natura Ekos Commands Price Premium



Natura Distributes Value to Align Incentives



Note: Data from Natura's Annual Report 2019 and Annual Report 2016. GRI 203-1 indicator.



Ronald H. Coase

The Problem of Social Cost

*“Pigou’s conclusion and that of most economists using standard economic theory was, and perhaps still is, that **some kind of government action** (usually the imposition of taxes) was required to restrain those whose actions had harmful effects on others, often termed **negative externalities**.*

*What I showed in that article, as I thought, was that in a **regime of zero transaction costs, an assumption of standard economic theory, negotiations between the parties would lead to those arrangements being made which would maximise wealth and this irrespective of the initial assignment of rights.***

-- Ronald H. Coase, [Nobel Prize Lecture](#)

All good, but ...

**...how do we
measure the
impact?**



Satellite Monitoring System

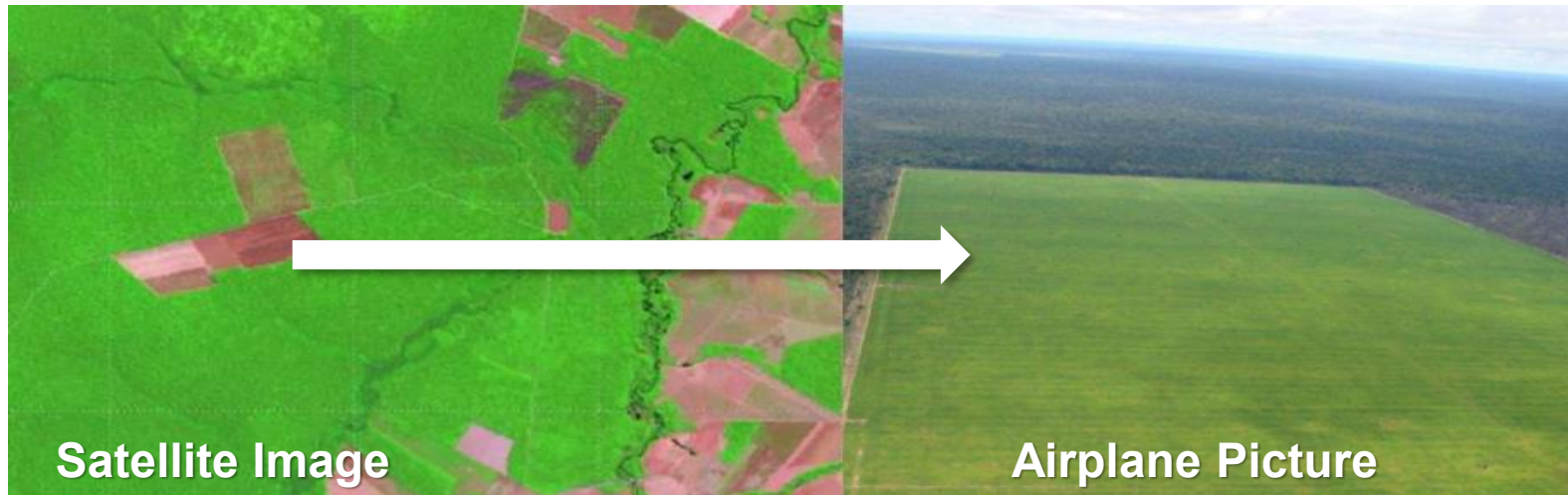
Images **Landsat** (NASA/USA) spatial resolution 30 meters;

Monitoring system has **international reliability** ([Kintisch, 2007- Science](#)).

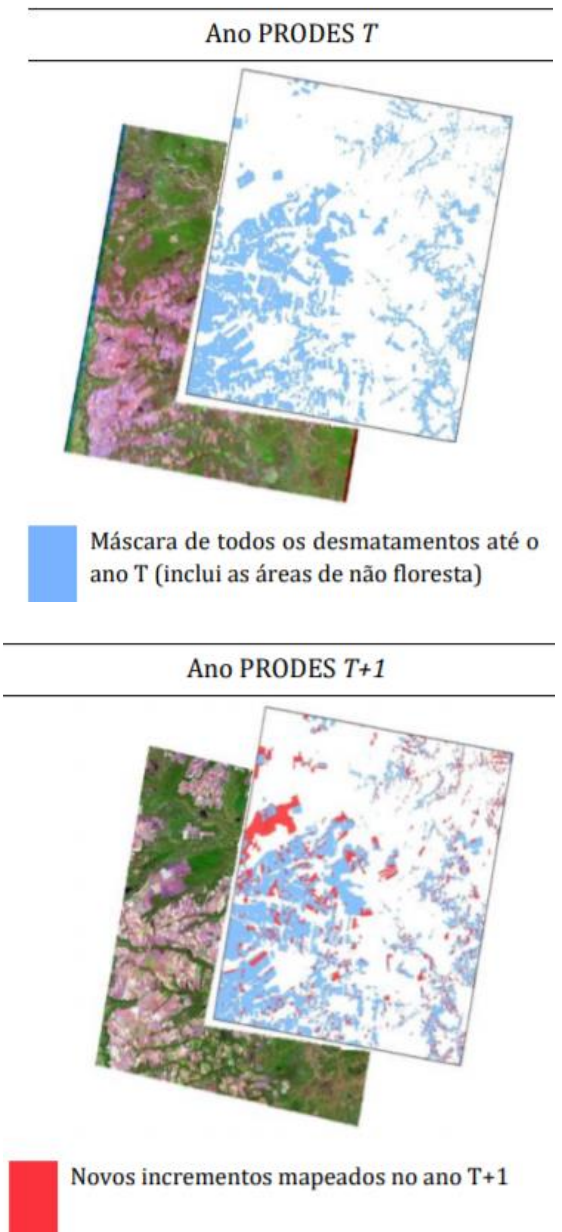
Detect **deforestation of shallow cut**;

i.e., total deforestation in a window of time;

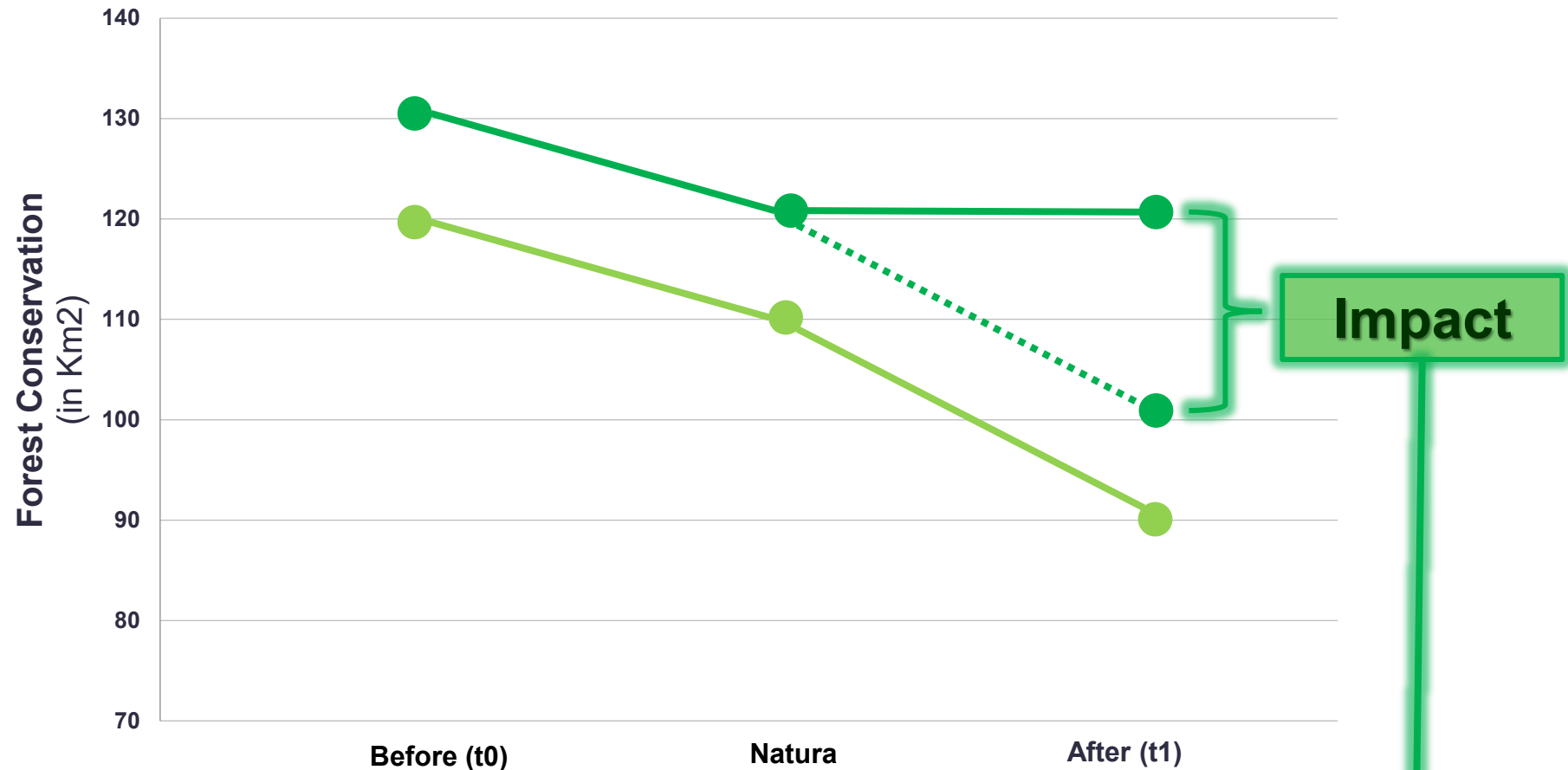
analyze the yearly increment of deforestation;



Source: PRODES. Available at < http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/apresentacao_prodes-1.pdf > And < http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes/pdfs/Metodologia_Prodes_Deter_revisada.pdf > Accessed on October 22nd, 2019



Additionality Principle



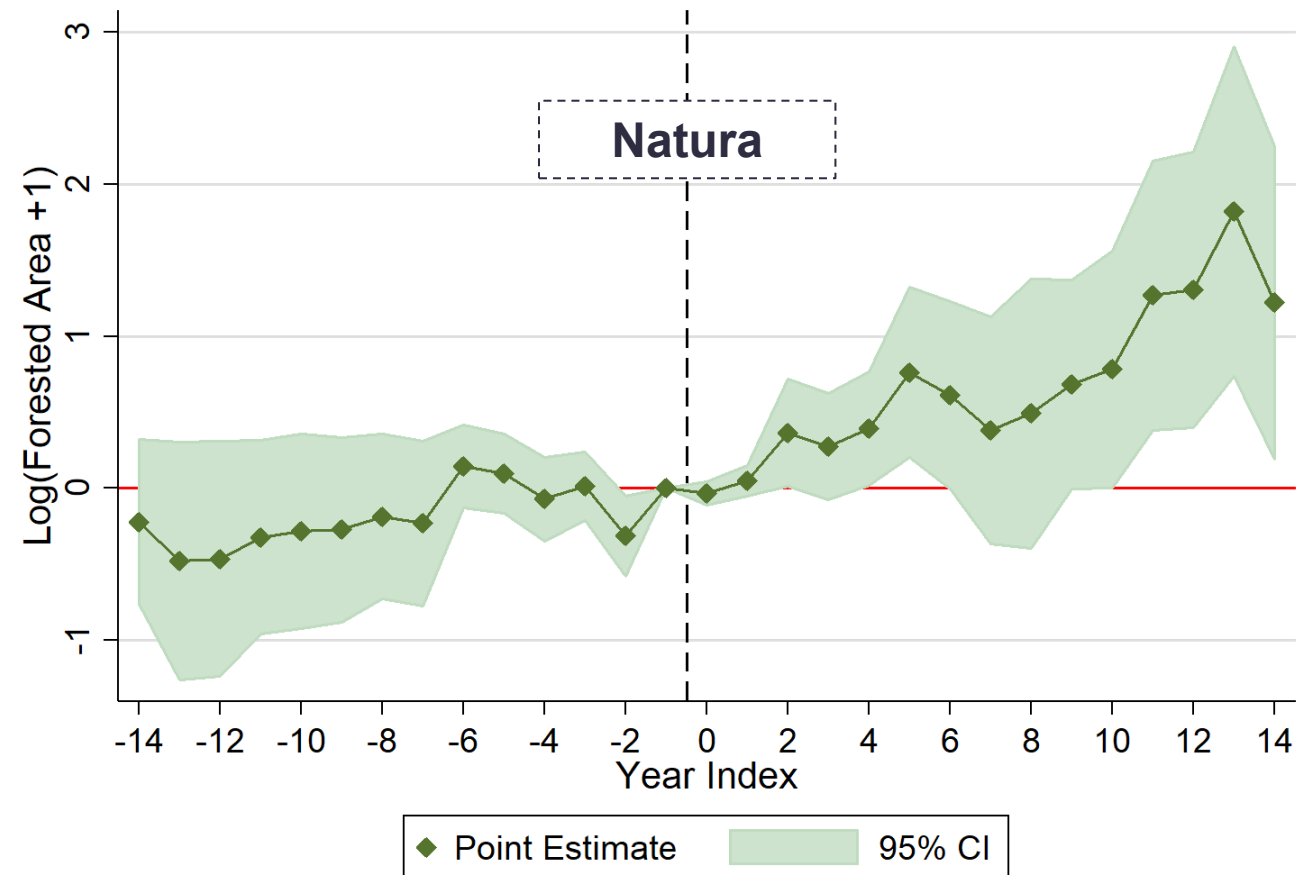
$$\begin{aligned} \text{IMPACT} &= \\ &= [\text{Natura in } t1 - \text{Natura in } t0] - [\text{NonNatura in } t1 - \text{NonNatura in } t0] \end{aligned}$$

	(1)	(2)
DV = $\text{Log}(\text{Forested Area} + 1)$		
Natura	0.468* (0.231)	0.479* (0.224)
Year FE	Y	Y
Municipality FE	Y	Y
Satellite Controls	N	Y
Constant	5.232** (0.0371)	5.303** (0.0368)
Observations	14,440	14,440
R-squared	0.949	0.951
Number of counties	760	760

Note: Robust standard errors clustered at the municipality level in parentheses.

** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

Increased Forest Preservation



Note: standard errors clustered at the municipality level.

On Average, Natura **increases** Forest Preservation by **47.9%** ($p < 0.05$)

1.8 million hectares ~ 1.7 million football fields

~ 145 million tons of Carbon.

Missing Satellite Images

Missing Images are usually associated with **technical problems** with the **satellite** at the moment of picture collection.



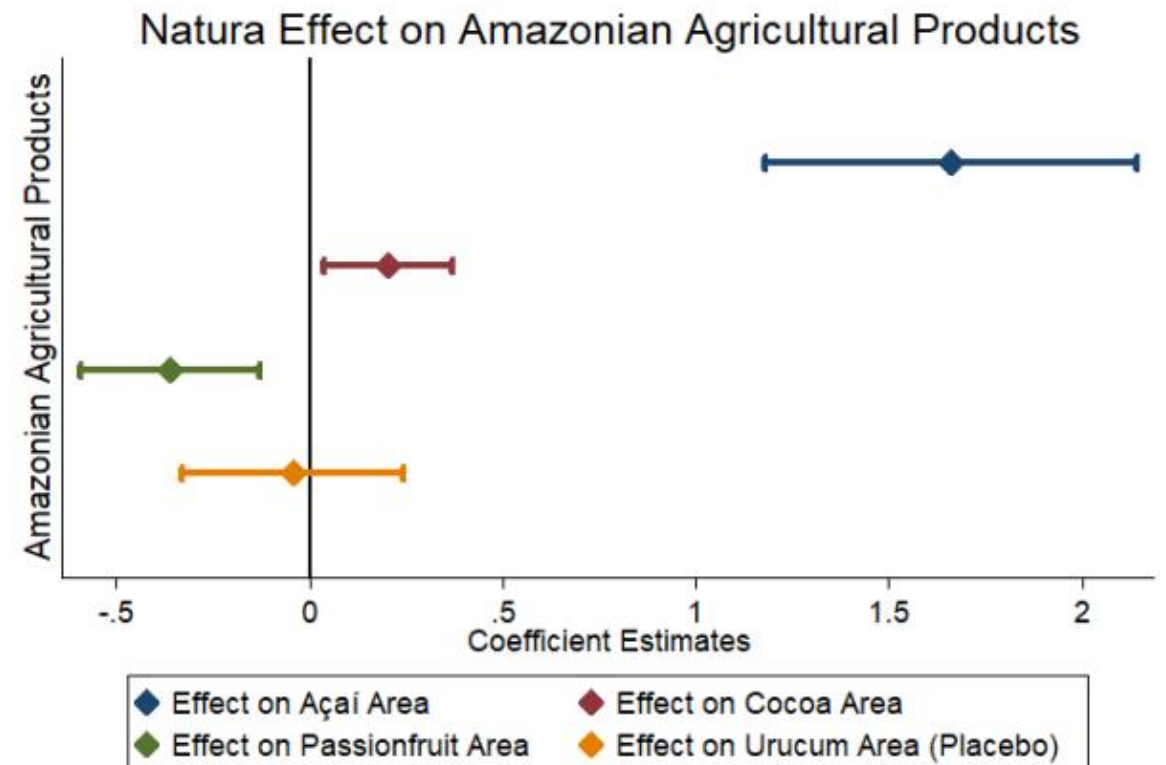
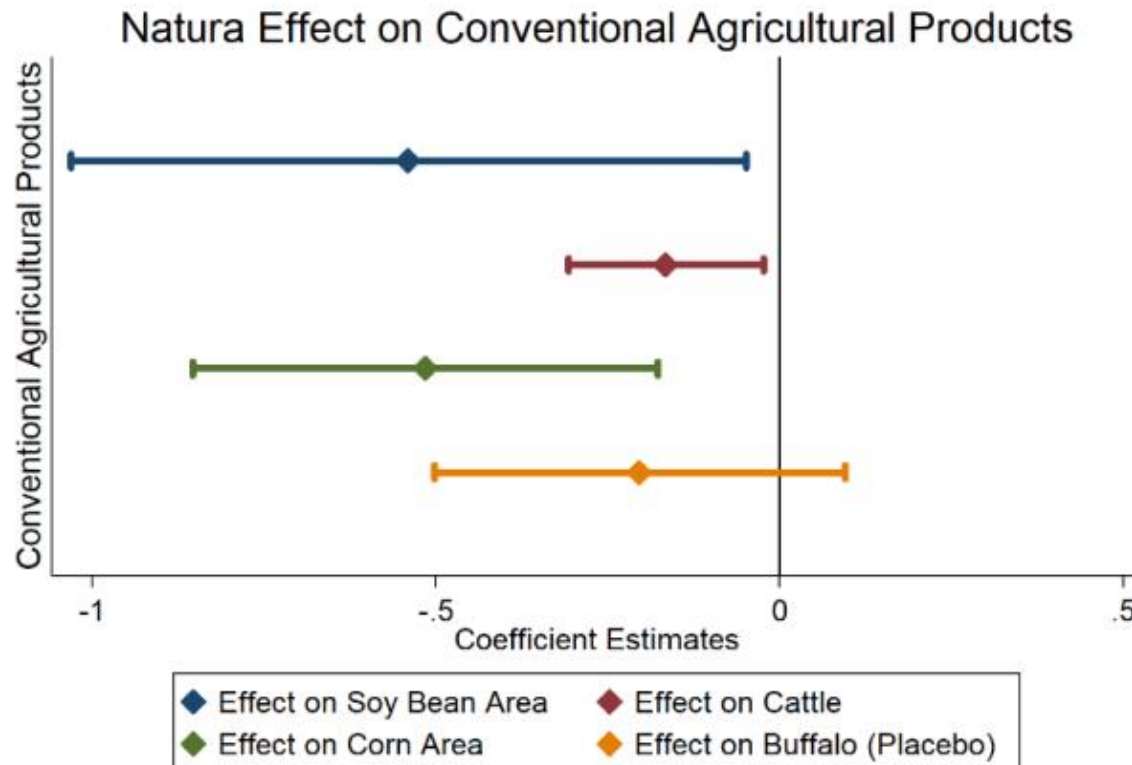
	(1)	(2)
DV =	Second Stage: Log(Forested Area+1)	First Stage: Natura
Natura	19.36** (5.374)	
Log(Missing Image +1)t-1 × (1 - Natura)t-1		-0.0127** (0.00361)
Year FE	Y	Y
Municipality FE	Y	Y
Satellite controls	Y	Y
R-squared	0.552	0.704
Observations		13,676
Number of municipalities		760
Weak Identification test F-stat		12.46

Note: Robust standard errors clustered at the municipality level in parentheses.
 ** p<0.01, * p<0.05, + p<0.1

- **Relevance:** Natura managers affirm using satellite data to evaluate the production potential of the municipality before entry.
- **Independence:** Technical problems in the satellite are independent of topographic coverage.
- **Exclusion Restriction:** Loggers and farmers do not use this information on their decision to deforest.

Conventional vs. Amazonian Agriculture

- Reduction in Conventional Agricultural Products, but an Increase in Amazonian Agricultural Products
- This leads to the increase in **Forested Areas**



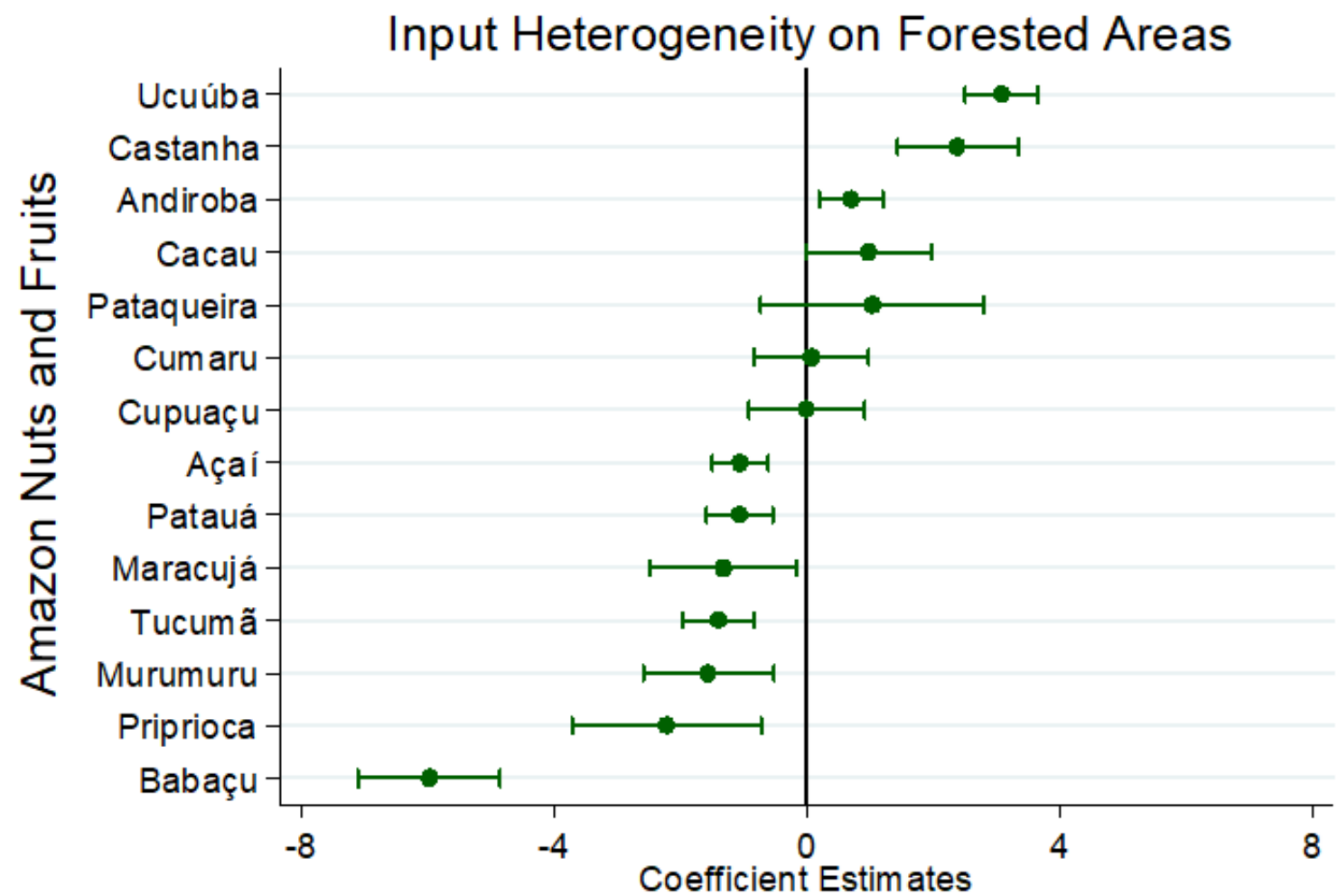
The power of incentive alignment with stakeholders

“If it wasn't for cocoa, you had to move cattle (...) with cocoa, a family that has 10 hectares of cocoa already supports themselves (...) this 10 hectares generate about R\$3.000 month.”

-- Reginaldo Silva, organic cocoa supplier.

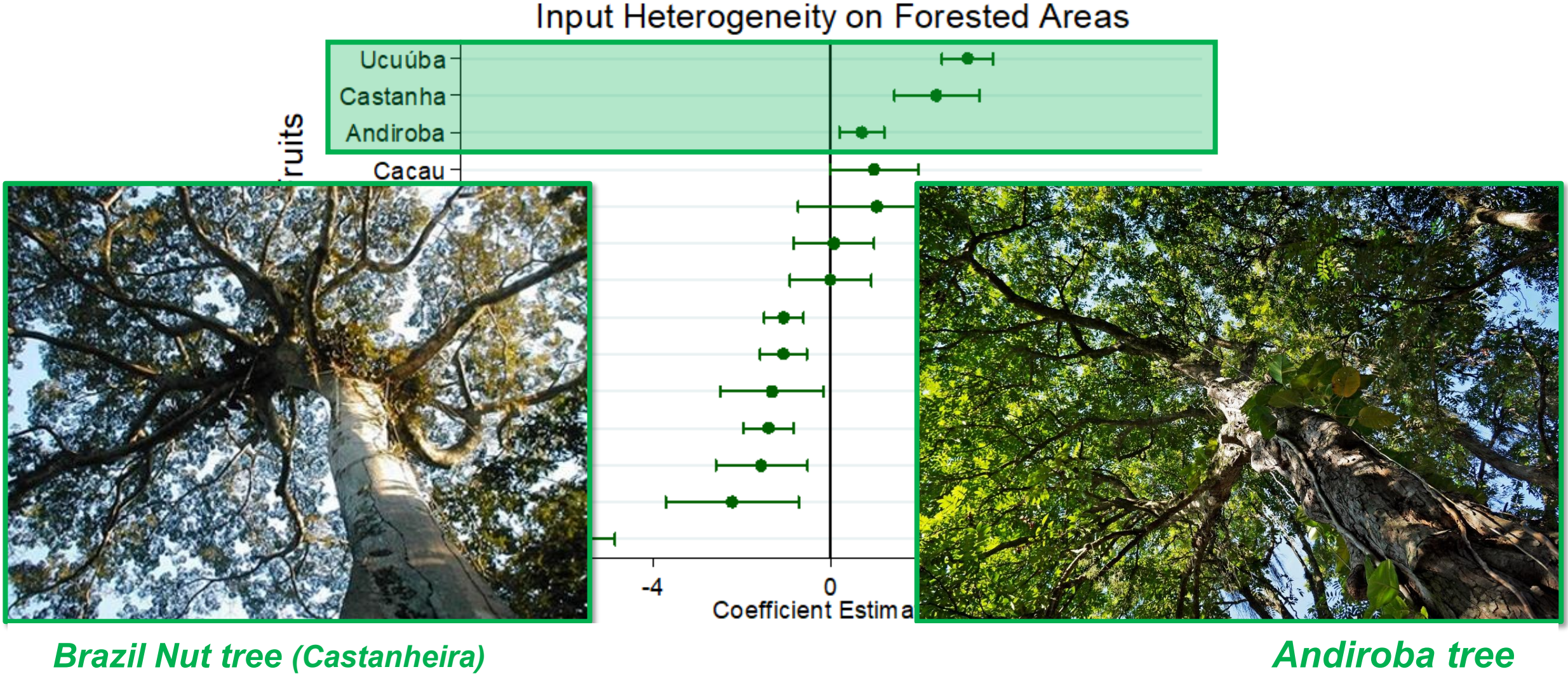


Effects on Forest Conservation and Regeneration



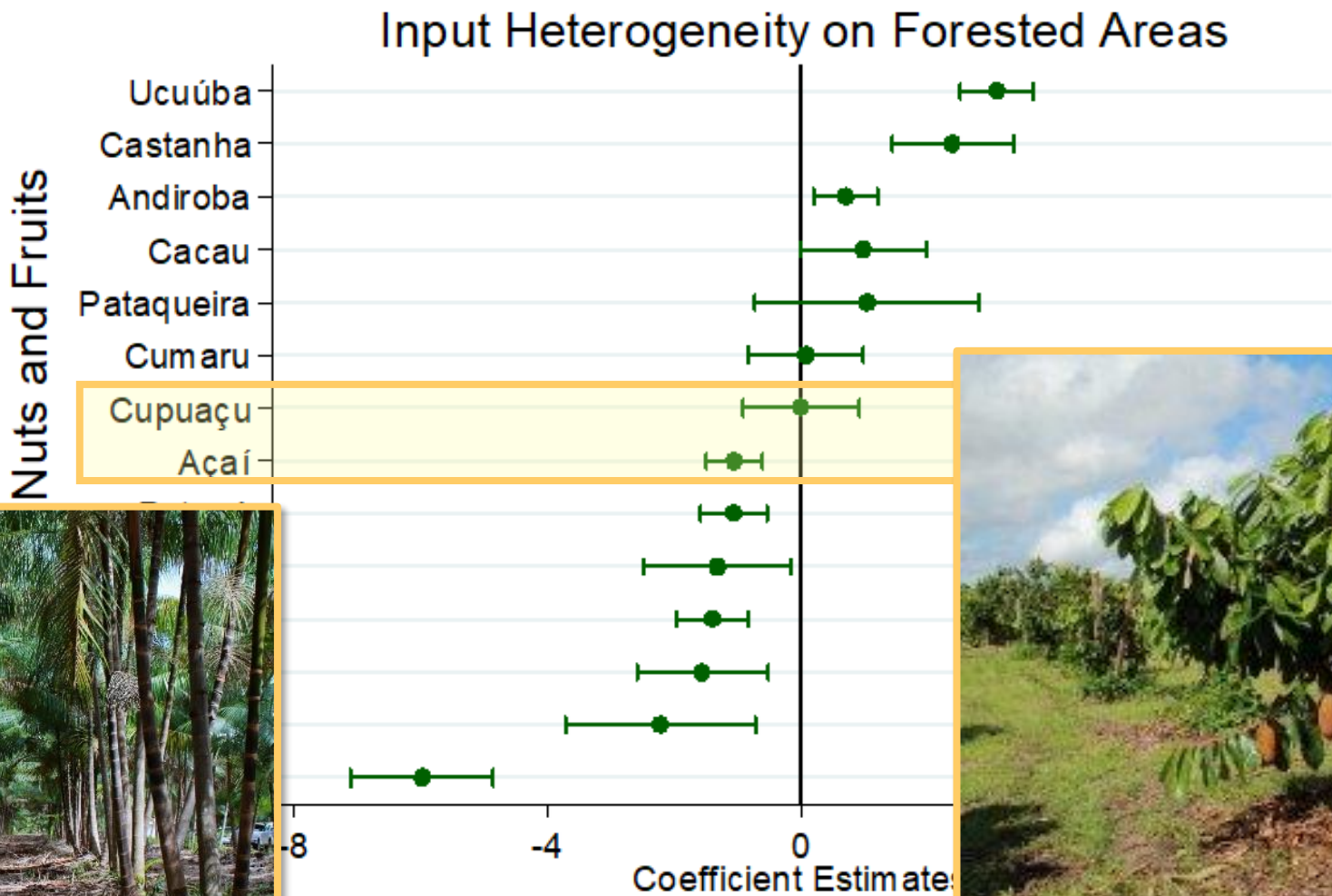
Note: Year FE's and robust standard errors.

Effects on Forest Conservation and Regeneration



Effects on Forest Conservation and Regeneration

Açaí palm



Cupuaçu trees



Note: Year FE's and robust standard errors.

Estimation of Natura's Effect in Carbon Tons

Estimated Effect of Natura in Carbon Tons saved and potential Price of the Positive Externality

Carbon stock in Amazon (tC/ha)	- 2sd	- 1sd	Mean	+ 1sd	+ 2sd
	26.4	53.5	80.6	107.7	134.8

Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura Estimated conservation takes into account the total number of hectares “saved” by Natura (total of 1.807.640 hectares), which are based on the estimations presented in the previous slide.

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

Estimation of Natura's Effect in Carbon Tons

Estimated Effect of Natura in Carbon Tons saved and potential Price of the Positive Externality

Carbon stock in Amazon (tC/ha)		- 2sd	- 1sd	Mean	+ 1sd	+ 2sd
		26.4	53.5	80.6	107.7	134.8
Natura Estimated Conservation in Amazon (in tC)		47,721,702	96,708,752	145,695,802	194,682,852	243,669,903
		Natura Estimated Conservation in (millions €/tC)				
Price of Carbon under the European Union Emissions Trading System (€/tC)	€ 5	€ 239	€ 484	€ 728	€ 973	€ 1,218
	€ 15	€ 716	€ 1,451	€ 2,185	€ 2,920	€ 3,655
	€ 25	€ 1,193	€ 2,418	€ 3,642	€ 4,867	€ 6,092
	€ 35	€ 1,670	€ 3,385	€ 5,099	€ 6,814	€ 8,528

Note 1: Carbon stock in Amazon considers the difference between forested and deforested areas. Source (Souza-Rodrigues, 2018: 2730).

Note 2: Natura Estimated conservation takes into account the total number of hectares "saved" by Natura (total of 1.807.640 hectares), which are based on the estimations presented in the previous slide.

Note 3: Carbon prices set considering historical prices and the price on January 1st, 2020, which equals € 24.32/tC. Source: Bayer and Aklin (2020) and ICE EUA Futures. Available at: <<https://www.theice.com/products/197/EUA-Futures/data?marketId=5115271&span=3>> Accessed on April 16th, 2020.

Flashcards

First Sustainability-Linked Bond Focused on Biodiversity: R\$1.3B Raised by Natura

Natura issued the first sustainability-linked bond explicitly tied to biodiversity metrics, raising R\$1.3 billion (supported by IDB Invest). The bond's terms are linked to Natura's targets for expanding bio-ingredient sourcing in the Amazon. Separately, IFC (World Bank Group) made its first-ever investment in Amazon community farmers through Natura's Amazônia Viva financing mechanism (initial R\$6M investment via FUNBIO). Natura has pledged \$100M to find regenerative ways of cultivating trees, nuts, berries and plants used in its products.

Sources: [IDB Invest \(2024\)](#), [IFC Press Release \(2024\)](#), and [Mongabay: Financial Tools for Amazon Bioeconomy \(Dec 2025\)](#)

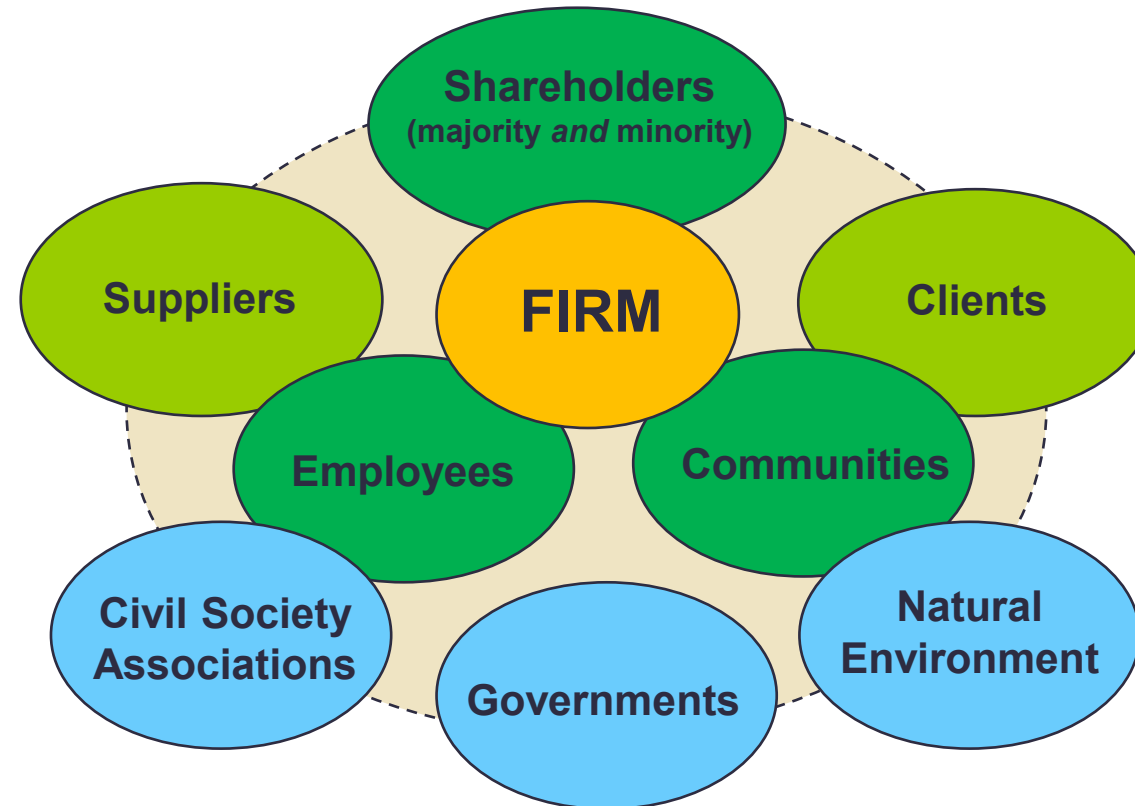
Question: “Our DID research shows Natura preserves 1.8M hectares worth \$728M–\$5.0B in carbon. Natura just raised R\$1.3B in a biodiversity bond. Is the market finally pricing the externality?”

ANALYSIS

Coasean Solution in Practice: The biodiversity bond is a Coasean solution. It create a market for the externality (biodiversity preservation) and private actors will internalize it. Investors now have a financial instrument whose returns depend on Natura sourcing more Amazonian bio-ingredients. **The externality becomes an investment thesis.**

Connecting DID Analysis to Finance: The R\$1.3B bond is only possible *because* of scientific evidence. Our findings provides the evidentiary foundation that makes biodiversity a credible investment. **Without measurement, the bond is just a green label. With measurement, it's a priced externality. This is why impact measurement matters: it can unlocks capital.**

The Stakeholder Management Framework



BUT, sometimes firms might want to mislead shareholders and other stakeholders

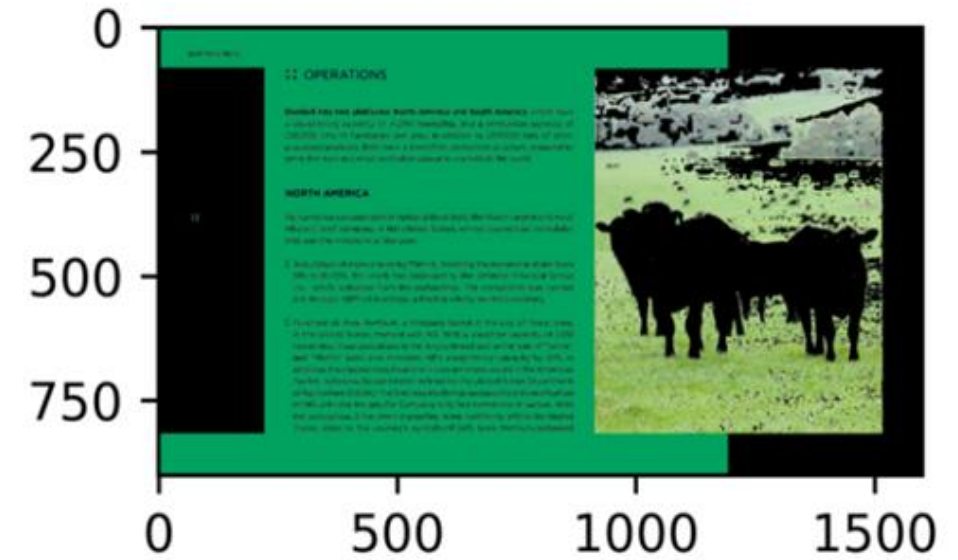
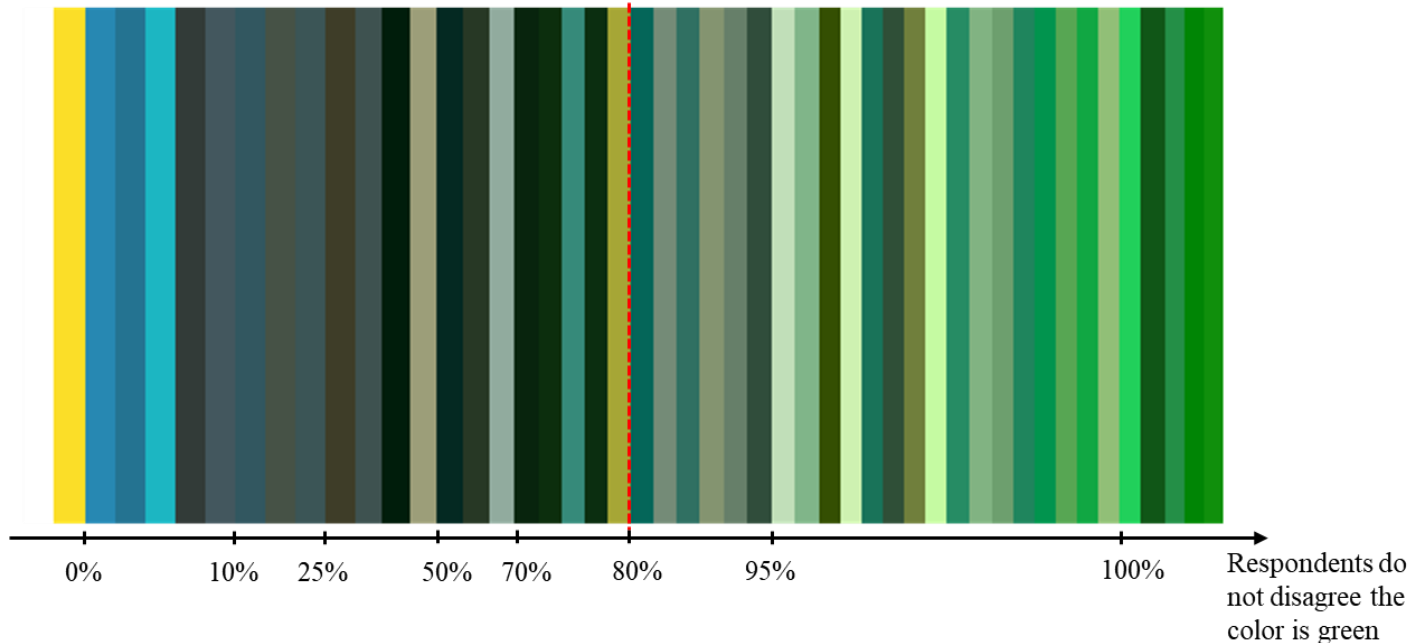
“Greenwashing”

A new toolkit to measure “greenwashing”

- **Textual Analysis** (using annual/sustainability reports)
 - Why? There is evidence that firms use language to obfuscate environmental performance mandated by standardized reporting
 - We create a dictionary based on the “Corporate Social Responsibility Words-Environment” dictionary (CAT Scanner)
 - We expand this “Envirolex” dictionary. Now it contains 321 terms, including Amazon specific ones (e.g., Amazon biome, Amazon rainforest, etc.)
 - We count the frequency of Envirolex words
- **Sentiment Analysis**
 - Use the Loughran and McDonald (L&M) dictionary (H4N) to measure sentiment
 - We count positive/negative terms in the report and in sentences that have Envirolex words
- **AI Visual Analysis** (AI image analysis on how green a report is)

Visual analysis

- We developed an **AI algorithm** to identify the number of **pixels green** in a firm's report
- **But, what is green?**
 - MTurk survey with 1,000 respondents helped us to define what the human eye considers green.
 - Define green using max/min boundaries of the HSV (Hue, Saturation, Value) space that comes out of our survey.



Results

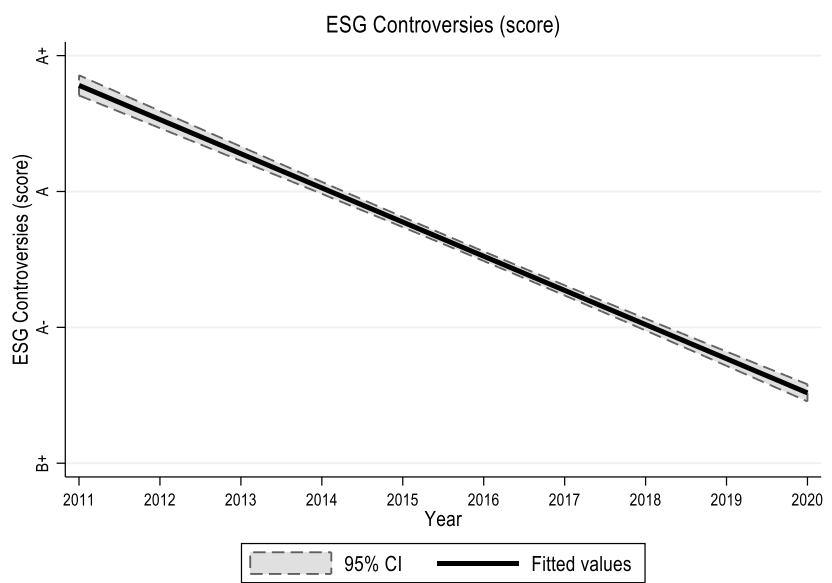
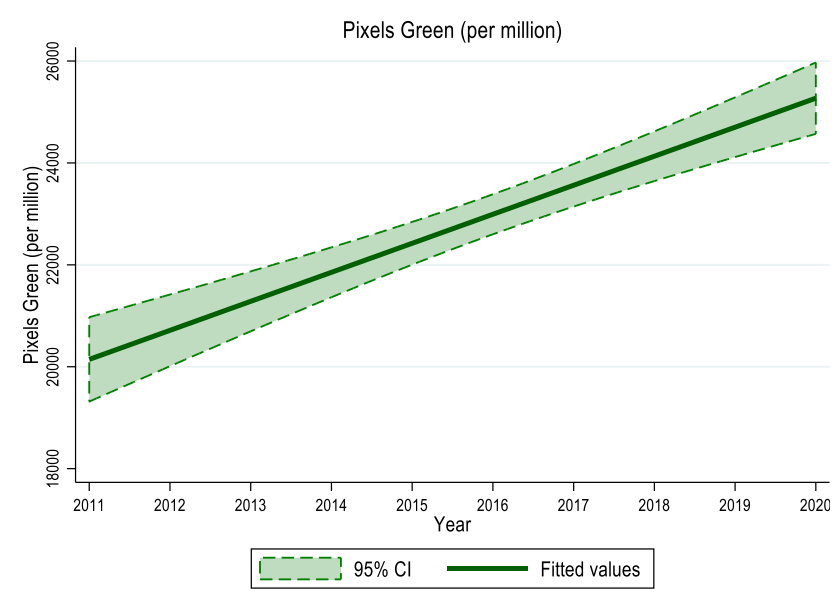
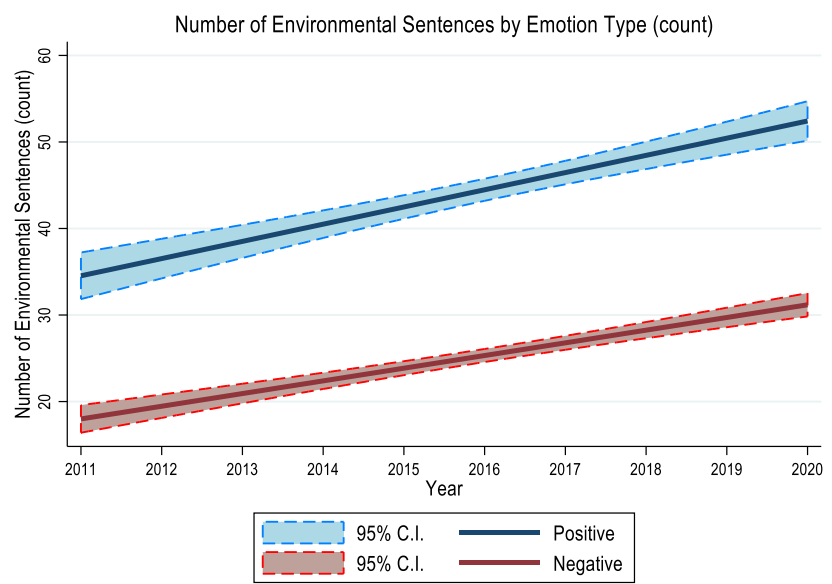
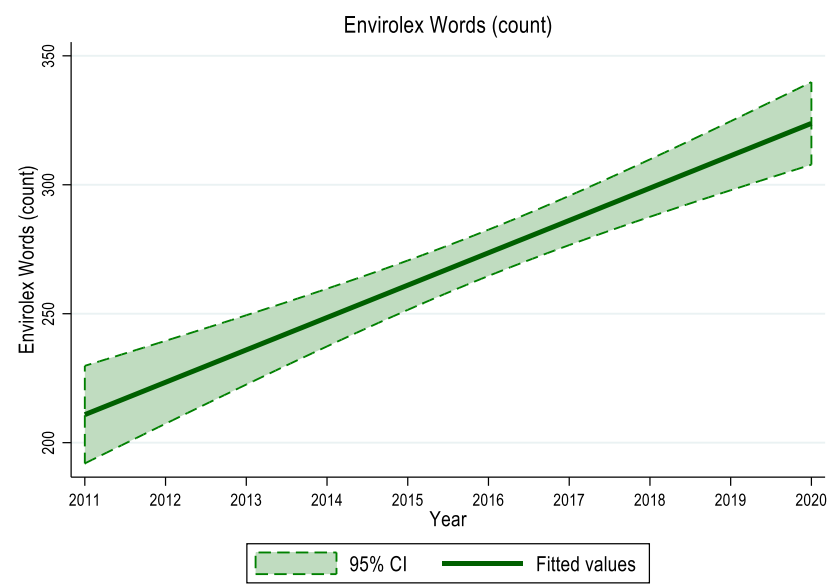
Over ten years

- 118 more environment-related words ($p = 0.000$)
- 15 additional environmental sentences with positive sentiment ($p = 0.000$)
- 16 additional environmental sentences with negative sentiment ($p = 0.000$)
- 5k additional pixels green per million ($p = 0.017$). 4k additional pixels green per report.

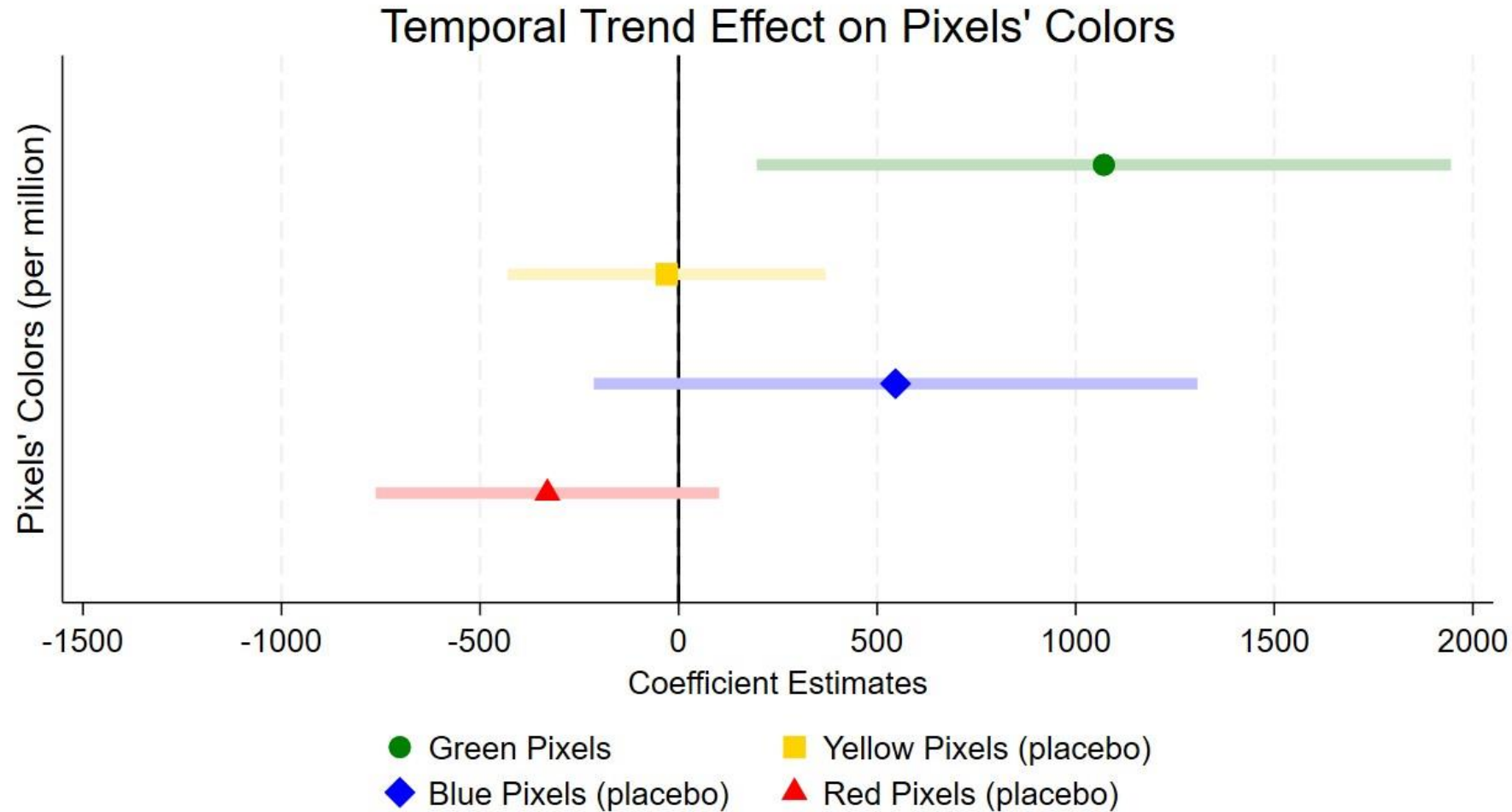
BUT...

- 2 points lower ESG Controversies score. That is, more fines related to environ. wrongdoing ($p = 0.004$)

Note: ESG controversies score from Thomson-Refinitiv database rates firms according to the number of environmental penalties and wrongdoings the firm has in a year.



Is it only green or all colors?



Note: Controlling for total page counts, year and firm FE's. Robust standard errors clustered at the year-sector level.

A falsification test using the primary colors (i.e., yellow, blue, and red) shows that the intertemporal increase is only significant in green pixels.

Flashcards

EU Bans Vague Environmental Claims from September 2026; Greenwashing Lawsuits Accelerate

The EU's Empowering Consumers Directive (EmpCo) and Environmental Claims Directive (ECD) take effect in 2026, **prohibiting unsubstantiated claims like “eco-friendly” or “green” without evidence**, unreliable sustainability labels, and false claims about product sustainability. Canada's Competition Act now requires substantiation for all environmental claims with steep penalties. Greenwashing class-action lawsuits are expected to surge across energy, aviation, retail, and financial services.

Sources: [Forward Law Review \(Jan 2026\)](#) and [Crowell & Moring: ESG Law Trends 2026](#).

Question: “From September 2026, the EU bans saying ‘eco-friendly’ without proof. Does regulation solve the problem, or will firms find new ways to greenwash?”

ANALYSIS

Greenwashing Detection, Institutional Catch-Up: Our research documented the problem: firms systematically increase green language and green visuals. The EU directive is the institutional response. This is the classic North (1991) pattern: **informal norms of greenwashing persist until formal rules constrain them.**

Stakeholder and Impact Measurement: Natura's model, grounded in measurable impact (DID, satellite data, carbon tons), is the best defense against greenwashing accusations. **If your stakeholder strategy is real and measurable, you can substantiate it under any regulatory regime.** The firms at risk are those with communication strategies disconnected from operations.

“The SDGs offer the potential to unite the most powerful organizations in the world – governments, corporations, and civil society – behind a single agenda to save humanity and the planet from suffering and devastation.

Including the private sector in their development and execution can greatly magnify the odds of success.”

(Kramer, Agarwal, & Srinivas, 2019: 9)

SUSTAINABLE DEVELOPMENT GOALS

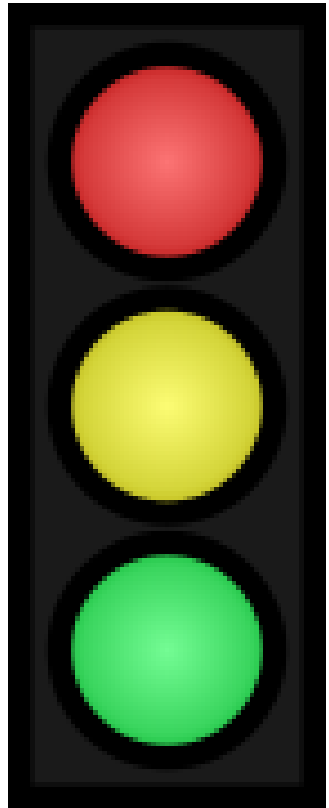


[This Photo](#) by Unknown Author is licensed under [CC BY](#)

Key Takeaways

- How should we go? **Stakeholder Management Framework**
 - An adequate stakeholder management can be an effective way by which firms (MNCs) could contribute to solve many global problems.
 - The internalization of externalities is a clever way by which firms can align economic, social, and ecological incentives.
 - However, some firms might be strategic in their disclosure aiming to mislead their shareholders and other stakeholders.
 - The additionality principle enables us to measure potential socioenvironmental impact and verify whether the firm is causing this impact or not.
 - The UN-SDGs provide a roadmap on which goals firms might contribute to achieve by 2030.

What We Are Trying to Solve in the Global Module



Should we go abroad?

- **ADDING Value**
- **CIA:** Comparative Institutional Analysis

- Ferrero Group
- Corruption in International Business

Where should we go?

- **CAGE:** Culture, Administrative, Geographic, and Economic
- **AAA:** Adaptation, Aggregation, and Arbitrage

- European Super League
- Apple
- Zara

How should we go?

- **TCE:** Transaction Cost Economics
- **SM:** Stakeholders Management

- Walmart
- McDonald's
- Tata Group
- Natura



Looking Ahead: Group Presentations

- Quiz opens at **5PM today** and closes in **72 hours**
 - **15 minutes to answer 3 questions** (open book)
 - Once you start the quiz, you can't stop or start over
- **Next Class:**
 - Class Presentations: Monday and Wednesday
 - Upload slides until April 20th by 8 am
 - Final paper due April 23rd by noon

Wrapping up the Global module...

What comes to your mind when you think about "Global Management"?



Time for the Global Module Evaluation



Thank you. Please stay in touch!

Leo Pongeluppe
pongelup@wharton.upenn.edu

